SHALIAH FRICAS

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EDUCATION

University of Massachusetts, Boston | Boston, MA Expected Graduation: May 2027
Bachelor of Science in Computer Engineering | Minor in Computer Science

Relevant Coursework: Python, Programming in C++, Data Structures & Algorithms, Circuit Analysis, Calculus 1-3, Honors Applied AI, Engineering Design, Discrete Math, Linear Algebra, Digital Systems, Physics
Certifications & Programs: TRIO McNair Scholars Program - Undergraduate Research Fellow, Agentic AI for Business, Multi-Tier AI Agent Architecture, Oxford Saïd Business School Artificial Intelligence Programme Alumni

RESEARCH

McNair Scholar - ICARUS Lab | (Mentor: Prof. Zografopulos) | UMass Boston February 2026 - Present
Undergraduate Researcher - Robotics & Cyber-Physical Systems
- Utilizing ICARUS Lab research capabilities to analyze how cyber-physical attacks influence robot behavior, system reliability, and human trust in assistive technologies - furthering research in Masters Program.

UCaN Lab | (Advisor: Prof. Rahaim) | UMass Boston
Undergraduate Researcher September 2025 - Present
- Investigating hybrid RF and optical wireless communication systems using real-time hardware and live data, supporting applications in indoor sensing, localization, and reliable data transmission.

ENGINEERING PROJECTS

Interactive Robotic Companion | ESP32, Sensors, Displays, Embedded C++, Robotics | *In Progress*
- Designing and building an interactive robotic companion using ESP32 - integrating sensors, display output, and embedded firmware to enable responsive behavior and human-robot interaction (Adventure Time Inspired)

AFRL/ HyLo SDR Research Project | SDR, MATLAB, Python, RF, OWC, Raspberry Pi | *In-Progress* | Link
- **AFRL Finalists:** Developing a hybrid indoor localization system that combines RF fingerprinting with optical wireless communication (OWC/VLP) to improve accuracy in environments where GPS fails.

MediSeek (Biomedical AI Troubleshooting Agent) | AI Agent Editor, Generative AI | Link
- Designed an AI-powered troubleshooting agent that supports biomedical engineers and healthcare professionals in diagnosing medical device faults, improving response time, and reliability in clinical environments

WORK EXPERIENCES

Operation Autopilot, Inc. | Boston, MA
Robotics Field Concierge Intern (Hybrid) January 2026 - Present
- Supporting live deployments of autonomous cleaning robots by diagnosing navigation, task execution, and docking issues, while coordinating field operations and communicating system health to customers and teams

NeuralSeek | Miami, FL
AI Agent Builder Intern (Remote) August 2025 - September 2025
- Collaborated with a team to design and deploy generative AI agents that address real-world challenges for industries like HR, Healthcare, and Business

Arka HR | Boston, MA
Operations and App Development Intern June 2023 - December 2024
- Streamlined internal HR workflows by organizing data, coordinating across teams, and supporting process improvements, gaining hands-on experience with structured operations and cross-functional collaboration.

TECHNICAL SKILLS

Skills: Cyber-Physical systems, PC Building, IT Troubleshooting, Device Repair, Digital/Analog Circuits, C++, Python, Optical Wireless Communication/Li-Fi, Operational Reliability, Autonomous Systems, Robotic Operations
Technologies: GNU Radio, LTspice, MATLAB, VS Code, KiCad, Raspberry Pi, Matplotlib, CAD, Arduino IDE, Opal RT, Microsoft Office (Word, PowerPoint, Excel), Google-Suite (Doc, Slides, Sheets), Canva, Adobe

ADDITIONAL SKILLS

Languages: Bilingual in English and Spanish